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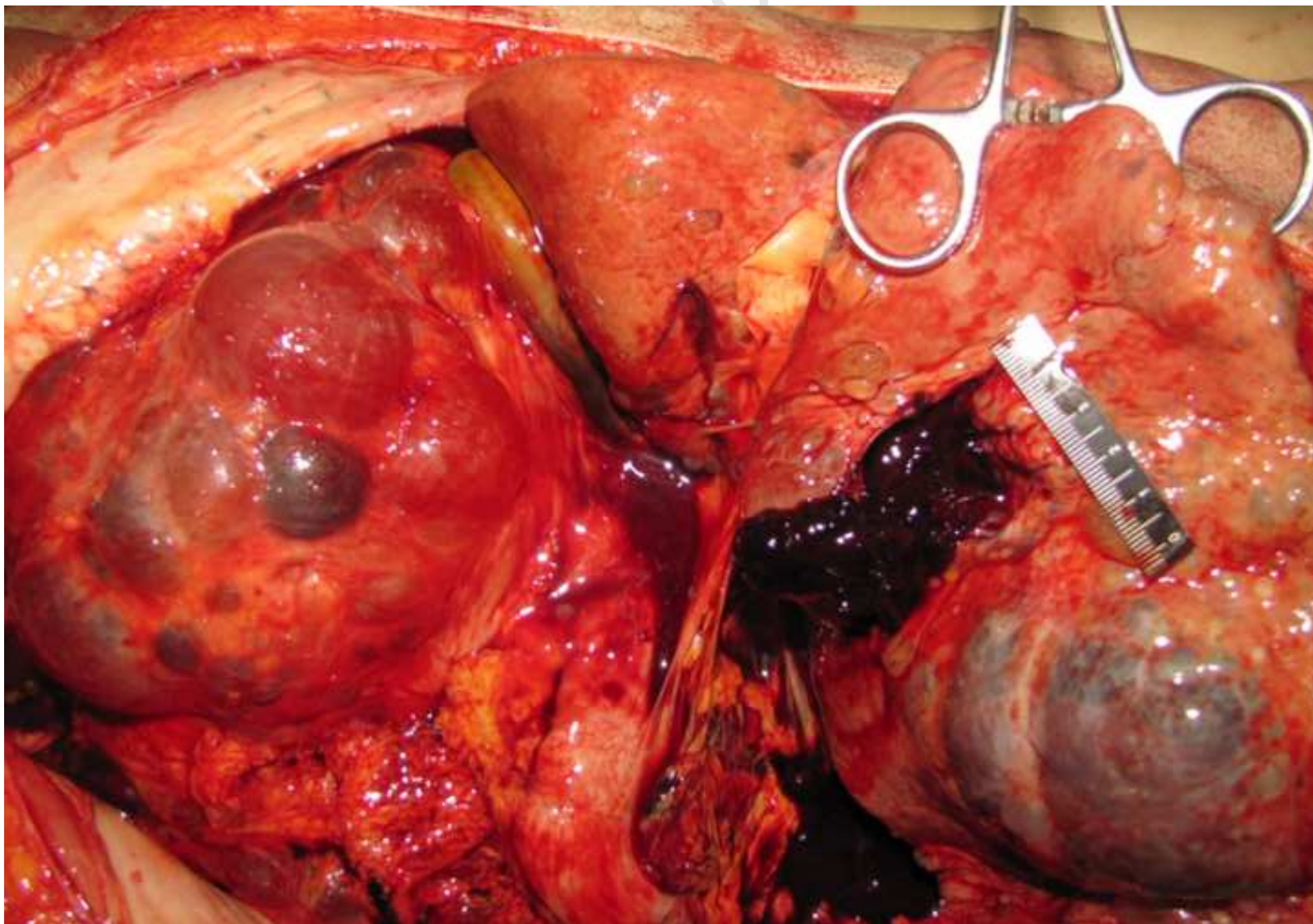
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Highlights

1. There are only a few reports of fatal cases about PLD and ADPKD.
2. Analysis of a complex relationship among trauma, disease and death is involved.
3. Genetic analysis plays an important role in the diagnosis of PLD.
4. All the data in this report are consistent with the literature.

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Fatal Liver Cyst Rupture in Polycystic Liver Disease Complicated with Autosomal Dominant Polycystic Kidney Disease: A Case Report

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Fatal Liver Cyst Rupture in Polycystic Liver Disease Complicated with Autosomal Dominant Polycystic Kidney Disease: A Case Report

Abstract

A 59-year-old man was struck in the abdomen and later presented to the emergency room. His blood pressure dropped and eventually died 16 hours post trauma and just before emergency exploratory laparotomy. Autopsy revealed two polycystic kidneys and a giant polycystic liver with two ruptures. Blood (2225g) was observed in the peritoneum and the body-surface injury was minor. Genetic testing was performed to confirm that the man had an autosomal dominant polycystic kidney disease (ADPKD) complicated by polycystic liver disease (PLD). Autopsy, histopathology and medical history showed that the cause of death was the ruptures of liver cysts due to trauma. In this communication, we describe a fatal case and hope to increase awareness and recognition of PLD and ADPKD. We also wish to indicate that due to the fragile condition of liver cysts, trauma should be considered even if the body-surface injury is minor in fatal cases of PLD patient with a traumatic history.

Key words: Forensic pathology, autosomal dominant polycystic kidney disease, polycystic liver disease, genetic analysis.

Introduction

Polycystic Liver Disease (PLD) is a rare anomaly caused by malformation of biliary duct in the liver [1-3]. The disease was first reported in 1856 in combination with autosomal dominant polycystic kidney disease (ADPKD) [4]. There are three kinds of PLD entities: ADPKD, autosomal dominant polycystic liver disease (PCLD) and Von Meyenburg complexes [5, 6], all of which have different pathologies and pathogenesis. Investigators have shown that PLD is clinically silent in early stage and approximately 50% of PLD patients with massive hepatomegaly will suffer from severe complications such as cyst hemorrhage, cyst rupture, and/or a cyst infection in the final stages [4, 7, 8].

Autopsy and histopathology findings in this particular case showed that the PLD of the deceased was associated finding of ADPKD. ADPKD is a life-threatening disease characterized by fluid-filled renal cysts, which could lead to renal failure due to the progressive enlargement of cysts [9]. Complications such as PLD, intracranial aneurysms, aortic root dilatation and aneurysms, mitral valve prolapse, hypertension and abdominal wall hernias are well described in literature [3]. Genetically, a certain mutation in PKD1 and/or PKD2 is associated with ADPKD [3, 9].

Here, we present a case of a PLD patient who died following an impact in the abdomen that disrupted the liver cysts. A postmortem genetic analysis was performed to further confirm the diagnosis of ADPKD. To a certain extent, the minor body-surface injury of the deceased complicated the evaluation of the disease, trauma and death.

Case Report

A 59 year-old man was kicked in the abdomen in a fight at 12:00 noon. He fell to the ground and was sent to the emergency room two hours later. His blood pressure dropped to 75/65mmHg at 14:20. Blood work revealed WBC at $11.8 \times 10^9/L$, RBC at $3.38 \times 10^{12}/L$ and HB 98g/L. Ultrasonography showed significant fluid accumulation in

the peritoneum. He received a conservative treatment initially and was administered medications to maintain a normal blood pressure. At 2:00 a.m. of the next day (14 hours post-trauma) his blood pressure suddenly dropped to 85/52mmHg and abdominal distention was quite obvious. Abdominal tenderness and positive ascites signs were evident. It was decided that the patient required an emergency exploratory laparotomy to stop abdominal bleeding. At 2:52 a.m. (15 hours post-trauma), his heart stopped just before the surgery. Ultimately, he was announced dead at 4:17 a.m. despite all treatments. Past medical history revealed that the deceased had a history of chronic renal failure and hypertension and once had an operation to repair an abdominal wall hernia.

A forensic autopsy was performed two days after death. The body was 157.0cm long. The abdomen was obviously enlarged and there was an abdominal wall hernia, 6.5cm in diameter with a surgical scar right beside it. External examination did not show any lethal injury.

The liver was large, 3347g in weight and 32.0cm×21.0cm×12.0cm in volume and contained many cysts. The liver edge was extended below the costal margin and there was a large quantity of blood, 2255g, mainly beneath the liver in the peritoneum. The surface of the liver was irregular, with many cysts throughout. The cysts varied in size, were thin-walled and filled with either brown or yellow fluid. We found a giant ruptured cyst, 11.0cm×7.0cm×5.0cm in size, on the left lobe of the liver with an 11.0cm long rupture on the visceral surface and a large volume of clotted blood adhering around the rupture (fig.1a. and 1b.). There were smaller ruptured cyst, 5.0cm×3.0cm×2.5cm, near the boundary of the two lobes. The length of the second rupture was 2.5cm, and with parenchymal hemorrhage. Microscopically, the cysts were lined with columnar, cuboidal or flattened epithelium and surrounded by a scant of fibrous tissue with infiltration of inflammatory cells. Hemorrhage was observed in the tissue around the rupture (fig.2.). Most of the parenchyma was normal.

The two kidneys were both irregularly shaped with many cysts (fig.3.). The left one was 650g in weight and 15.0cm×9.0cm×8.5cm in size, and the right, 2000g in weight and 22.0cm×12.0cm×11.0cm in size. Histopathological examination showed

numerous cysts surrounded by cuboidal epithelium and hyperplasia of fibrous connective tissue in the renal interstitial tissue (fig.4.).

The heart was 610g in weight and coronary atherosclerosis was observed in the LAD, LC and RM. The most severe atheromatous plaque was in the LAD, occupying more than 75% of the lumen. Microscopic examination showed the left ventricle and interventricular septum had fibrosis.

Most of the main organs of the deceased were anemic in appearance. No other significant pathological changes were found.

Genetic Tests

DNA sequencing library was prepared. Briefly, genomic DNA was extracted from EDTA-anticoagulated peripheral blood and broken up by ultrasonication. Subsequently, a chip was prepared with target DNA and subjected to high throughput sequencing to screen for mutations.

Analysis showed that the deceased had a mutation of PKD1 (NM_001009944.2), which was found to be a splice heterozygous for a nucleotide mutation in intron 20, c.7863+2T>G (fig.5.).

Diagnosis

Autopsy and clinical records indicated that the cause of death was rupture of liver cysts. Genetic tests revealed a heterozygous mutation and further confirmed the diagnosis of ADPKD.

Discussion

ADPKD is a potentially lethal monogenic disorder with a prevalence between 1 in 400 to 1 in 1000 in all races worldwide [3]. The presence of fluid-filled renal cysts along with cysts in other organs, such as liver and pancreas, is the most characteristic feature of ADPKD [10]. Renal failure happens when normal kidney tissues are substituted by a large number of cysts [3], and at this stage, the average decline rate of

GFR will be about 4.4-5.9 mL/min/year [11]. In this particular case, the patient had a

medical history of chronic renal failure but the renal function was, however, maintained by medical treatment.

The pathogenesis of ADPKD is related to mutations in two genes: PKD1 and PKD2 which account for approximately 85% and 15% of all such cases, respectively [12-15]. Mutation in either gene can give rise to the disease [9]. In clinical practice, imaging examination, such as CT, MRI and ultrasonography are common methods in the diagnosis of ADPKD and genetic testing could be used for further confirmation [3]. In the case presented here, postmortem genetic analysis revealed a heterozygous mutation in PKD1: c.7863+2T>G, which the literature describes as one of the causes of ADPKD.

PLD is the most common complication of ADPKD [16]. The typical pathological changes of the PLD liver are numerous thin-walled cysts, which contain clear or light yellow fluid without bile, surrounded by columnar, cuboidal or flattened epithelium [17]. In the case described here, changes in the liver and kidneys agreed with the PLD complicated with ADPKD. ADPKD patients also show connective tissue defects such as abdominal wall hernias that happen to 15-45% of ADPKD patients due to the high volumes of the liver and kidney [3, 18]. The deceased here had a 4.0cm×3.0cm abdominal wall hernia. Hypertension is another common symptom in ADPKD [19]. The heart weight of the deceased was 610g, which was a sign of hypertension.

Hemorrhage and infection due to spontaneous rupture of cysts are severe complications of PLD [20]. In the case described here, ruptures might have been initially mistaken as a complication of PLD because the body-surface injury was too minor to support it as traumatic. However, according to the medical history, hemorrhage was closely associated with the trauma, evidenced by the distended abdomen and the continuously deteriorating condition of the patient. Besides, the size of the liver was 32.0cm×21.0cm×12.0cm, much larger than normal, resulting in

hepatomegaly with the edge extending past the costal margin. Due to the medical condition of the patient, he was vulnerable to trauma. Thus, we concluded that it was the force of impact in the abdomen that disrupted the thinly walled cysts and led to hemoperitoneum.

In forensic pathology, PLD and ADPKD are occasionally encountered. Detailed autopsy and histopathology examination should be performed to make a tentative diagnosis. Since, to the best of our knowledge, there are three kinds of pathogenesis which can give rise to PLD [3], genetic analysis is helpful in the diagnosis of the specific etiology. A comprehensive review of the medical history is needed in order to evaluate the relationship among disease, trauma and death. Due to the fragile condition of the liver in PLD patients, trauma should be considered in the first place even if the body-surface injury is minor.

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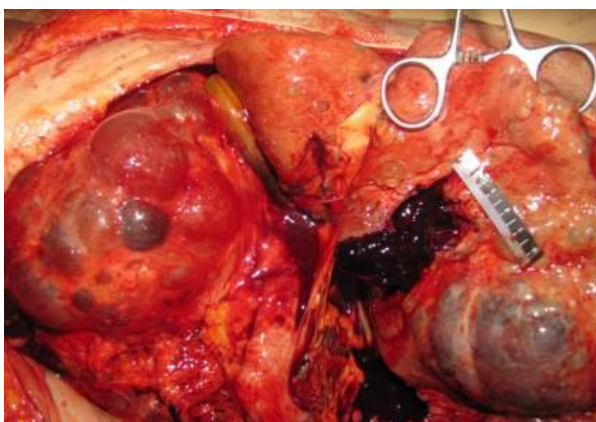


Fig.1a. The rupture on left lobe of the liver with clotted blood around it.



Fig.1b. The rupture on left lobe of the liver with clotted blood around it.

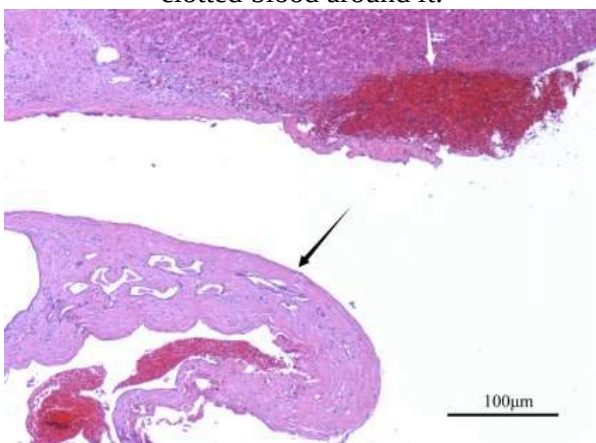


Fig.2. The clotted blood and rupture of the liver (white arrow) , liver cyst (black arrow)
HE $\times 40$

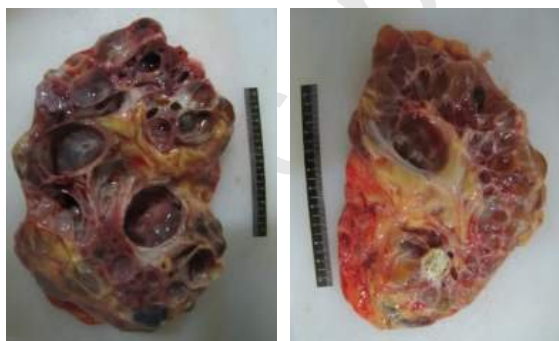


Fig.3. The two kidneys were both irregularly shaped with many cysts.

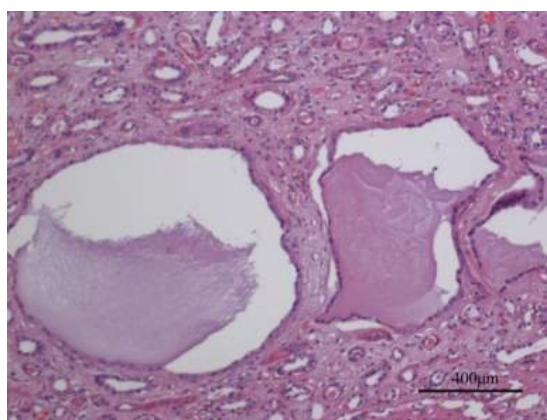


Fig.4. Kidney cysts on microscope HE $\times 100$

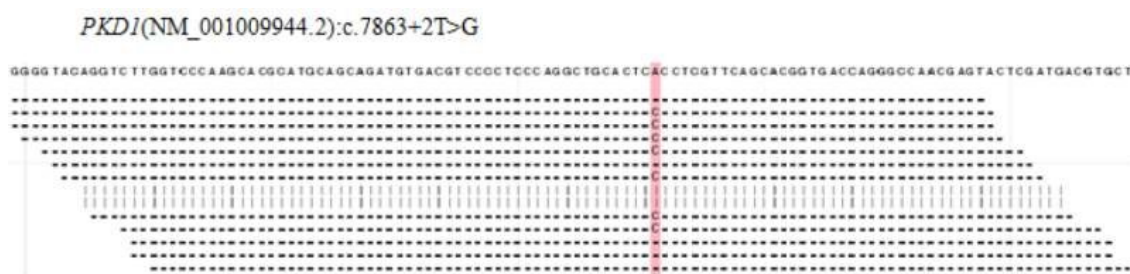


Fig.5. Genetic tests

The picture above is the result of plus-strand reads of the detection area. The highlighted area is the mutation of c.7863+2T>G.

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